Priyanka Sinha

Optimization, Machine Learning, Generative Modeling, Game Theory
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Education

- **North Carolina State University** — Raleigh, NC
  Ph.D. in Electrical Engineering; Minor in Mathematics; — *January 2018 - October 2022*
- **University of Florida** — Gainesville, FL
  Master of Science in Electrical Engineering; — *August 2011 - May 2013*
- **National Institute of Technology** — Durgapur, West Bengal, India
  Bachelor of Technology in Electrical Engineering; — *July 2006 - September 2010*

Work Experience

- **University of Maryland, Baltimore County, Mathematics and Statistics** — Baltimore, MD
  Postdoctoral Researcher — *August 2026 - Current*
  - Research on diffusion generative models for medical imaging, federated learning, distributed optimization, and inverse problems.
- **Virginia Tech, Math Department** — Blacksburg, VA
  Departmental Postdoctoral Fellow — *2024 - August 2026*
  - Research on large scale inverse problems, distributed optimization, and federated learning.
- **Virginia Tech, ECE Dept** — Arlington, VA
  Presidential Postdoctoral Fellow — *November 2022 - December 2023*
  - Research on application of game dynamics on Federated Learning, Adversarial Learning and Watermarking.
- **NCSU, ECE Dept** — Raleigh, NC
  Research Assistant, MPACT Lab — *January 2018 - October 2022*
  - Research on application of signal processing, machine learning, and probability theory for the detection, localization, and tracking of aerial vehicles.
- **Qualcomm Inc** — San Jose, CA
  Senior WLAN Systems Engineer — *June 2013 - November 2017*
  - Developed beamforming and dynamic frequency selection (DFS) algorithms for IEEE 802.11ac WiFi chips.

Projects

- **Steering and Diffusion Representation for Medical Imaging**: We study whether pretrained diffusion models encode longitudinal disease progression in a form that can be separated from patient specific anatomical variation and used for early change detection without lesion localization. We propose a patient centered framework based on differences between baseline and follow up diffusion activations, with Recursive Feature Machines and Average Gradient Outer Product geometry used to extract low dimensional progression coordinates for sequential monitoring with empirically calibrated false alarm control. We also examine rare and spatially localized pathology through the same representation perspective, asking when clinically important structure is preserved or lost during diffusion learning and reverse sampling.
- **Stable Collaboration in Heterogeneous Federated Learning**: Developed a framework for learning collaboration structure among heterogeneous federated clients, where a single global model may not serve all clients equally well. The method uses client performance feedback to infer compatibility among clients and form stable collaboration groups with lower internal distributional variation. Client similarities are updated through no regret learning, and aggregation weights are adapted based on observed cross client performance rather than fixed averaging rules. The resulting framework connects fairness, communication efficiency, and model stability by allowing clients with compatible objectives to share information more strongly while reducing harmful aggregation across incompatible clients.
- **Collapse and Recovery in Clustered Federated Learning**: Studied initialization induced cluster collapse in clustered federated learning as a controlled dynamical system, with cluster models as the evolving state and client routing as the control input. Characterized a starvation basin in a two group linear model and derived finite horizon recovery conditions based on persistent selective exposure, requiring sufficient compatible updates to a disadvantaged model while controlling incompatible contributions. Developed client side blocked EXP3 routing controllers that recover distinct usable specialists in linear and Gaussian mixture settings, while transformed FEMNIST experiments expose a coordination limit in which slower neural specialization requires coordinated and persistent commitment across clients. Submitted to the DynaFront 2 workshop at NeurIPS 2026.
- **Distributed Dynamic Inverse Problems**: Designed a decentralized Consensus-ADMM framework for total variation regularized tomographic reconstruction over a network of nodes, each with different heterogeneous observability. Introduced edge weights and coordinate wise masks that encode this heterogeneous coverage, and consensus is enforced only where nodes carry reliable information about each pixel. Developed multiplicative weight updates for these masks with regret guarantees, yielding adaptive communication patterns that approach the best fixed consensus structure in hindsight while scaling to large image ensembles. .
- **Dynamic Coreset Selection for Large Scale Machine Learning Models** : Developed a dynamic coreset selection framework that adapts to model evolution during training, using multilinear extension and continuous greedy optimization to preserve gradient accuracy while greatly reducing data usage.

Publications

- Workshop Publications
  - **P. Sinha** and S. Chatterjee, "The Dynamics of Collapse and Recovery in Clustered Federated Learning," accepted to appear in *DynaFront 2: Dynamics at the Frontiers of Optimization, Sampling, and Games, NeurIPS 2026 Workshop*, December 2026.
  - **P. Sinha** and S. Chatterjee, "Targeted Diffusion Models for High Generative Accuracy of Low-Probability Events," abstract accepted for poster presentation at the *New York Scientific Data Summit (NYSDS) 2026*, 2026.
- Conference Publications
  - **P. Sinha**, J. Kibilda and W. Saad, "On the Tradeoff Between Heterogeneity and Communication Complexity in Federated Learning," in *Proc. Asilomar (ACSSC)*, Pacific Grove, CA, USA, Oct. 29, 2023, 115-121, doi: 10.1109/IEEECONF59524.2023.10476887
  - **P. Sinha**, H. Krim and I. Güvenç, "Neural Network Based Tracking of Maneuvering Unmanned Aerial Vehicles," in *Proc. Asilomar (ACSSC)*, Pacific Grove, CA, USA, Nov. 3, 2022.
  - M. M. U. Chowdhury, **P. Sinha** and I. Güvenç, "Handover-Count based Velocity Estimation of Cellular-Connected UAVs," in *Proc. Int. Wrkshp. Signal. Proc. Adv. Wirel. Commun. (SPAWC)*, Atlanta, GA, USA, May. 2020.
  - **P. Sinha**, Y. Yapici, I. Guvenc, E. Turgut, and M. C. Gursoy, "RSS-Based Detection of Drones in the Presence of RF Interferers," in *Proc. IEEE Consumer Communications & Networking Conference (CCNC)*, Las Vegas, USA, Jan. 10-13, 2020, pp. 1–6.
  - **P. Sinha**, Y. Yapici, and I. Guvenc, "Impact of 3D Antenna Radiation Patterns on TDOA-Based Wireless Localization of UAVs," in *Proc. IEEE INFOCOM Workshops*, Paris, France, Apr. 29-May. 02, 2019, pp. 1–6.
  - J. Chen, Y. Zhou, D. Raye, W. Khawaja, **P. Sinha**, and I. Guvenc, "Impact of 3D UWB Antenna Radiation Pattern on Air-to-Ground Drone Connectivity," in *Proc. IEEE Veh. Technol. Conf. (VTC)*, Chicago, IL, USA, Aug. 2018.
- Journal Publications
  - **P. Sinha**, N. Alberti, S. Kang, M. Matthews, and G. Matthews, "Dynamic Coreset Selection of Large Scale Machine Learning," in preparation for *Jrnl. Machine. Learn. Res. (JMLR)*, 2026.
  - **P. Sinha**, and I. Guvenc, "Impact of Antenna Pattern on TOA Based 3D UAV Localization Using a Terrestrial Sensor Network," accepted to appear in *IEEE Trans. Veh. Technol.*, Feb. 2022.
  - **P. Sinha**, M. M. U. Chowdhury and I. Güvenç, David W. Matolak, and Kamesh Namuduri "Wireless Connectivity and Localization for Advanced Air Mobility Services," accepted to appear in *IEEE Aerosp. Electron. Syst. Magz.*, Feb. 2022.
  - **P. Sinha**, I. Guvenc and M. C. Gursoy, "Fundamental Limits on Detection of UAVs by Existing Terrestrial RF Networks," *IEEE Open J. Commun.* Soc., vol. 2, Sep. 2021, pp. 2111 - 2130.

Graduate Coursework

- **Mathematics**: Real Analysis, Convex Analysis, Optimization in Vector Spaces, Probability Theory, Stochastic Processes, Stochastic Differential Equations
- **Signal Processing**: Digital Signal Processing, Noise in Linear Systems, Detection and Estimation Theory
- **Machine Learning**: Pattern Recognition, Neural Networks

Other Skills

- **Programming Languages**: MATLAB, Simulink, Python

Service

- Co-organizer for the Applied Numerical Analysis Seminar at Virginia Tech, managing speaker invitations and seminar logistics.
- Reviewer: IEEE Transactions of Wireless Communications, IEEE Wireless Communications Letters, IEEE International Conference on Communications, Physical Communication, Computer Communications.
- Conference Travel Grant Chair at IEEE Dynamic Spectrum Access Networks (DySPAN) 2024.

Honors, Awards, and Outreach

- Presidential Postdoctoral Fellowship Award, Virginia Tech, 2022.
- Outstanding Achievement Award at University of Florida for the years 2011-12.
- Selected for Summer Internship in the Global Internship Program in Engineering Design and Innovation (GIPEDI), AICET, IIT Delhi, July, 2009.
- Graduated with distinction in Bachelors of Technology, at National Institute of Technology, Durgapur - September, 2010

References

- Gretchen Matthews, Professor (Email: gmatthews@vt.edu)
  Department of Mathematics, Virginia Tech, VA, Blacksburg, USA

- Ismail Guvenc, Associate Professor (Email: iguvenc@ncsu.edu)
  Electrical and Computer Engineering, North Carolina State University, NC, Raleigh, USA

- Mustafa Cenk Gursoy, Professor (Email: mcgursoy@syr.edu)
  Electrical and Computer Science, Syracuse University, Syracuse, NY, USA Electrical and Computer Engineering, North Carolina State University, NC, Raleigh, USA

- Hamid Krim, Professor (Email: ahk@ncsu.edu)
  Electrical and Computer Engineering, North Carolina State University, NC, Raleigh, USA

- Sayandev Mukherjee, Distinguished Technologist (Email: s.mukherjee@cablelabs.com)
  CableLabs, Santa Clara, CA, USA

- Susinder Rajan Gulasekaran, Director of Engineering (Email: susinder.gulasekaran@commscope.com)
  Senior IEEE Member, Ruckus Wireless, Sunnyvale, CA, USA

- Qinfang Sun, Senior Director (Email: qinfang@qti.qualcomm.com)
  Qualcomm, Inc., San Jose, CA, USA